



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Certified Lean Six Sigma Green Belt (CLSSGB) Training

TGS Ref No: TGS-2025055775

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 2

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1	1 June 2026	Initial release - CLSSGB lesson plan.	Dr. Alfred Ang
2	19 July 2026	Major revision: rebuilt as a 4-day Green Belt progression from the Yellow Belt (CLSSYB, TGS-2025053922). Content deepened to the CSSC Green Belt scope (Chapters 1-24) - adds sampling and sample size calculation, Measurement System Analysis and Gage R&R, hypothesis testing with p-values and Type I/II errors, correlation and regression, FMEA and RPN, Design of Experiments, SPC control chart selection with the eight out-of-control rules, and process capability Cp/Cpk. Expanded from 10 to 25 hands-on labs across the five DMAIC phases; teaching diagrams imported from the original v21 trainer deck; schedule and labs realigned to the v6 WA (SAQ) and PP assessment papers.	Dr. Alfred Ang

TABLE OF CONTENTS

Course Information.....	4
Learning Outcomes.....	4
Assessment.....	4
Course Schedule.....	5
Day 1 — Six Sigma foundations, the Green Belt role and the Define phase.....	5
Day 2 — Measure — process mapping, sampling, MSA and baseline capability.....	5
Day 3 — Analyze — root cause, hypothesis testing, correlation and regression.....	6
Day 4 — Improve, Control and the final assessment.....	7
Lab Reference (aligned to the DMAIC phases).....	8

Course Information

Course Title	Certified Lean Six Sigma Green Belt (CLSSGB) Training
WSQ Course Reference	TGS-2025055775
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	4 days · 8 training hours per day (32 hours)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on Lean Six Sigma labs using the Northwind Retail Distribution Centre improvement scenario
TSC Alignment	Quality Process Control (ELE-QUA-5006-1.1)
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Lead a Lean Six Sigma project — define the problem, scope the work and charter the project (A1, A2).
- LO2: Map and baseline a process using SIPOC, detailed process maps and value stream maps (A2).
- LO3: Build a valid measurement system — sampling, sample size and MSA/Gage R&R — and baseline capability (A4).
- LO4: Analyse process data statistically using Pareto, run charts, hypothesis testing, correlation and regression (A3).
- LO5: Identify and prove root causes of variation using Fishbone, 5 Whys and statistical evidence (A3, K2).
- LO6: Select, risk-assess and pilot improvements using solution selection matrices, FMEA and DOE (A5).
- LO7: Sustain the gain with SPC control charts, process capability and a control plan (A5, K1).

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 2 questions (K1, K2), 60 minutes, open book.
- Practical Performance (PP) — 3 applied DMAIC tasks (A1-A5), 90 minutes, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Final assessment is conducted on Day 4 from 4:00 pm.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Day 1 — Six Sigma foundations, the Green Belt role and the Define phase

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, course introduction, ground rules and mandatory digital attendance (AM) [slides 2–18]
10:00–11:15	75 min	FOUNDATIONS — What is Quality; Lean vs Six Sigma vs Lean Six Sigma; the five Lean principles; history of Six Sigma; $Y = f(X)$; classical vs Six Sigma quality [slides 19–57]
11:15–11:30	15 min	Tea break
11:30–12:30	60 min	FOUNDATIONS — sigma level and DPMO; the 1.5 sigma shift; Cost of Poor Quality; belt roles and what a Green Belt leads; project selection and the cause/solution matrix; the DMAIC roadmap and tollgates
12:30–13:30	60 min	Lunch break
13:30–15:00	90 min	Hands-on: Lab 1: The Green Belt Role, Project Selection and Business Case; Lab 2: $Y = f(X)$, Sigma Level, DPMO and the DMAIC Roadmap [slides 58–65]
15:00–15:15	15 min	Tea break
15:15–16:45	90 min	DMAIC · DEFINE — Voice of the Customer; affinity diagrams; Kano analysis; CTQ trees and operational definitions; the project charter; problem and goal statements; scope; SIPOC; stakeholders and RACI [slides 66–119]
16:45–18:00	75 min	Hands-on: Lab 3: Voice of the Customer, Affinity Diagram and Kano Analysis; Lab 4: CTQ Tree — Translating Customer Needs into Measurable Requirements [slides 120–127]
18:00–18:30	30 min	Day 1 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 2 — Measure — process mapping, sampling, MSA and baseline capability

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap and mandatory

9:45–11:00	75 min	digital attendance (AM) Hands-on: Lab 5: Project Charter, Problem Statement, Goal Statement and Scope; Lab 6: SIPOC, Stakeholder Analysis and RACI [slides 128–136]
11:00–11:15	15 min	Tea break
11:15–12:30	75 min	DMAIC · MEASURE — process mapping symbols and swimlanes; handoff analysis; the eight wastes (DOWNTIME); value-added analysis; value stream mapping; lean metrics and takt time [slides 137–211]
12:30–13:30	60 min	Lunch break
13:30–15:00	90 min	Hands-on: Lab 7: Detailed Process Mapping and Swimlane Analysis; Lab 8: Value Stream Mapping, Takt Time and the Eight Wastes [slides 212–220]
15:00–15:15	15 min	Tea break
15:15–16:30	75 min	DMAIC · MEASURE — types of data; data collection plans; sampling techniques; sample size calculation; Measurement System Analysis and Gage R&R; yield, DPU, DPO, DPMO and RTY; descriptive statistics and baseline capability
16:30–18:15	105 min	Hands-on: Lab 9: Data Types, Operational Definitions and the Data Collection Plan; Lab 10: Sampling Techniques and Sample Size Calculation; Lab 11: Measurement System Analysis and Gage R&R [slides 221–234]
18:15–18:30	15 min	Day 2 recap and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 3 — Analyze — root cause, hypothesis testing, correlation and regression

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 2 recap and mandatory digital attendance (AM)
9:45–11:00	75 min	Hands-on: Lab 12: Yield, DPU, DPO, DPMO, RTY and the Hidden Factory; Lab 13: Descriptive Statistics, Normality and Baseline Process Capability [slides 235–245]
11:00–11:15	15 min	Tea break
11:15–12:30	75 min	DMAIC · ANALYZE —

		common vs special cause variation; tampering; run charts and the six non-random patterns; the Pareto principle; stratification; boxplots [slides 246–316]
12:30–13:30	60 min	Lunch break
13:30–14:45	75 min	Hands-on: Lab 14: Variation, Run Charts and Stability Analysis; Lab 15: Pareto Analysis, Stratification and Boxplots [slides 317–326]
14:45–15:00	15 min	Tea break
15:00–16:30	90 min	DMAIC · ANALYZE — Fishbone (5M+E); 5 Whys; multi-voting; hypothesis testing (H_0/H_a, alpha, Type I and II errors, test selection, p-value decision rule); correlation and the coefficient of determination; regression
16:30–18:15	105 min	Hands-on: Lab 16: Fishbone, 5 Whys, Multi-Voting and Cause Prioritisation; Lab 17: Hypothesis Testing — Test Selection, p-values and Conclusions; Lab 18: Correlation, Regression and Quantifying the X-Y Relationship [slides 327–344]
18:15–18:30	15 min	Day 3 recap and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 4 — Improve, Control and the final assessment

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 3 recap and mandatory digital attendance (AM)
9:45–10:45	60 min	DMAIC · IMPROVE — solution generation and brainwriting; benchmarking; the solution selection matrix; 5S; poka-yoke; pull, JIT and Heijunka; standard work; FMEA and RPN; Design of Experiments; piloting [slides 345–391]
10:45–11:00	15 min	Tea break
11:00–12:30	90 min	Hands-on: Lab 19: Solution Generation, Benchmarking and Brainwriting; Lab 20: Lean Countermeasures — 5S, Poka-Yoke, Pull, JIT and Standard Work; Lab 21: Solution Selection Matrix and Cost-Benefit Analysis; Lab 22: FMEA, Risk Priority Numbers,

		DOE and Piloting [slides 392–413]
12:30–13:30	60 min	Lunch break
13:30–14:30	60 min	DMAIC · CONTROL — process capability Cp and Cpk; control limits vs specification limits; SPC and control chart selection; the eight out-of-control rules; the control plan; SOPs; visual management; verifying the gain and handover [slides 414–465]
14:30–15:15	45 min	Hands-on: Lab 23: Statistical Process Control — Chart Selection and Control Limits [slides 466–472]
15:15–15:30	15 min	Tea break
15:30–15:45	15 min	Hands-on: Lab 24: Control Plan, SOP, Visual Management and Response Plan; Lab 25: Verify the Gain, A3 Storyboard, Handover and Project Closure [slides 473–485]
15:45–16:00	15 min	Course recap and Briefing for Assessment [slides 486–497]
16:00–17:00	60 min	Written Assessment (WA) — Short-Answer Questions (SAQ), 2 questions, 60 minutes, open book
17:00–18:30	90 min	Practical Performance (PP) — 3 applied DMAIC tasks, 90 minutes, open book. PM digital attendance

Total training time: 480 minutes (8 hours).

Lab Reference (aligned to the DMAIC phases)

DMAIC phase / Topic	Weighting	Labs
FOUNDATIONS: Six Sigma Foundations & The Green Belt Role	10%	Lab 1, Lab 2
DEFINE: Define — Scope the Problem	18%	Lab 3, Lab 4, Lab 5, Lab 6
MEASURE: Measure — Quantify Performance	24%	Lab 7, Lab 8, Lab 9, Lab 10, Lab 11, Lab 12, Lab 13
ANALYZE: Analyze — Find and Prove the Root Cause	26%	Lab 14, Lab 15, Lab 16, Lab 17, Lab 18
IMPROVE: Improve — Select, De-Risk and Pilot the Fix	12%	Lab 19, Lab 20, Lab 21, Lab 22
CONTROL: Control — Hold the Gain	10%	Lab 23, Lab 24, Lab 25